source to drain through this channel. The gate terminal controls the flow of majority carriers from source to drain.

Working Principle of N-Channel JFET

Figure 6.10 shows the circuit of n-channel JFET with normal polarities. Here, the gate is kept negative potential with respect to the source called V_{GS}. The drain is kept positive potential with respect to the source known as V_{DS}. Let a voltage V_{DS} is applied across the drain and source terminals and no voltage is applied between gate to source (V_{GS}=*0*), now the two *PN* junctions establish a very thin and uniform depletion layer as shown in Figure 6.10(a). Thus, a large amount of electrons will flow from source to drain through a normally wide channel formed between the two depletion layers. This constitutes drain current I_D.

Let a reverse voltage V_{GS} is applied across the gate and source terminals (by closing the switch *S*) as shown in Figure 6.10(b). Now, the width of the depletion layer is increased due to the reverse biasing of the two *PN* junctions. This reduces the width of the conducting channel thereby decreasing the flow of electrons through it (I_D decreases). So the depletion layer will be thicker near the drain and the channel becomes wedge shaped as shown in Figure 6.10(b). Thus, the current flowing through the channel is controlled by width of the channel as well as reverse-potential applied to the gate. That is why the device is field effect transistor.

Advantages

1. The JFET has higher input impedance.
2. It is a low power consumption device.
3. It can be fabricated in small size area.
4. It has negative temperature coefficient of resistance, so they possess higher temperature stability.
5. It has less noise.

Disadvantages

1. The JFET is relatively low gain bandwidth product.
2. Its voltage gain is low.
3. It requires special handling during installation.

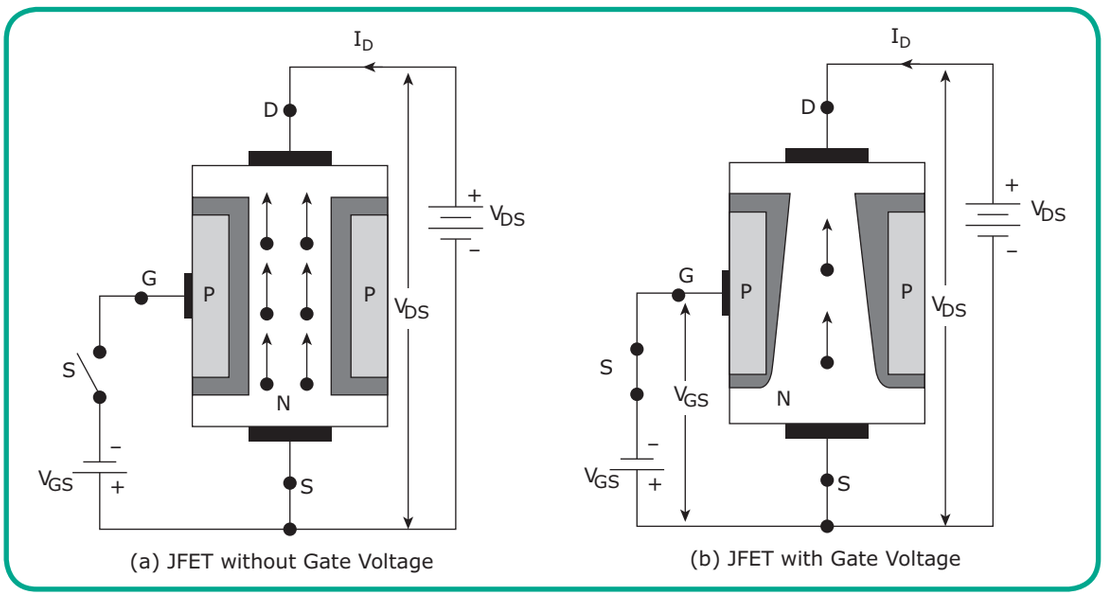


Figure 6.10 Working of N-Channel JFET with Normal Polarities

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